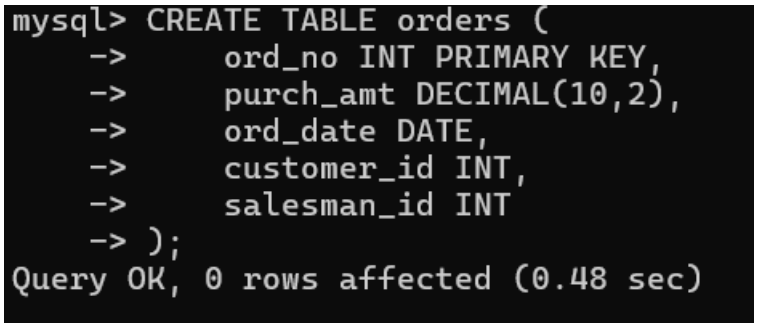
Skill-3

ORDERS TABLE

STEP 1: CREATE TABLE

```
CREATE TABLE orders (
    ord_no INT PRIMARY KEY,
    purch_amt DECIMAL(10,2),
    ord_date DATE,
    customer_id INT,
    salesman_id INT
);
```



STEP 2: INSERT DATA

```
INSERT INTO orders VALUES
(70001,150.50,'2012-10-05',3005,5002),
(70009,270.65,'2012-09-10',3001,5005),
(70002,65.26,'2012-10-05',3002,5001),
(70004,110.50,'2012-08-17',3009,5003),
(70007,948.50,'2012-09-10',3005,5002),
(70005,2400.60,'2012-07-27',3007,5001),
(70008,5760.00,'2012-09-10',3002,5001),
(70010,1983.43,'2012-10-10',3004,5006),
(70003,2480.40,'2012-10-10',3009,5003),
(70012,250.45,'2012-06-27',3008,5002),
(70011,75.29,'2012-08-17',3003,5007),
(70013,3045.60,'2012-04-25',3002,5001);
```

```
mysql> INSERT INTO orders VALUES
    -> (70001,150.50,'2012-10-05',3005,5002),
    -> (70009,270.65,'2012-09-10',3001,5005),
    -> (70002,65.26,'2012-10-05',3002,5001),
    -> (70004,110.50,'2012-08-17',3009,5003),
    -> (70007,948.50,'2012-09-10',3005,5002),
    -> (70005,2400.60,'2012-07-27',3007,5001),
    -> (70008,5760.00,'2012-09-10',3002,5001),
    -> (70010,1983.43,'2012-10-10',3004,5006),
    -> (70003,2480.40,'2012-10-10',3009,5003),
    -> (70012,250.45,'2012-06-27',3008,5002),
    -> (70011,75.29,'2012-08-17',3003,5007),
    -> (70013,3045.60,'2012-04-25',3002,5001);
Query OK, 12 rows affected (0.07 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

QUESTION 1 – WHERE CLAUSE

SCENARIO:
Finance wants all orders whose purchase amount is greater than $2,000.

QUERY:

```
SELECT *
FROM orders
WHERE purch_amt > 2000;
```

```
mysql> SELECT *
    -> FROM orders
    -> WHERE purch_amt > 2000;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
4 rows in set (0.04 sec)
```

QUESTION 2 – WHERE CLAUSE

SCENARIO:
Management wants all orders placed on 2012-09-10.

QUERY:

```
SELECT *
```

```
FROM orders
WHERE ord_date='2012-09-10';
```

```
mysql> SELECT *
    -> FROM orders
    -> WHERE ord_date='2012-09-10';
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
+--------+-----------+------------+-------------+-------------+
3 rows in set (0.01 sec)
```

QUESTION 3 - WHERE CLAUSE

SCENARIO:
Find all orders handled by salesman 5001.

QUERY:

```
SELECT *
FROM orders
WHERE salesman_id=5001;
```

```
mysql> SELECT *
    -> FROM orders
    -> WHERE salesman_id=5001;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
4 rows in set (0.01 sec)
```

QUESTION 4 - ORDER BY

SCENARIO:
CEO wants to see orders ranked from highest amount to lowest amount.

QUERY:

```
SELECT *
FROM orders
ORDER BY purch_amt DESC;
```

```
mysql> SELECT *
    -> FROM orders
    -> ORDER BY purch_amt DESC;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70010 |   1983.43 | 2012-10-10 |        3004 |        5006 |
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
|  70012 |    250.45 | 2012-06-27 |        3008 |        5002 |
|  70001 |    150.50 | 2012-10-05 |        3005 |        5002 |
|  70004 |    110.50 | 2012-08-17 |        3009 |        5003 |
|  70011 |     75.29 | 2012-08-17 |        3003 |        5007 |
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
12 rows in set (0.01 sec)
```

QUESTION 5 - ORDER BY

SCENARIO:
Operations team wants oldest orders first.

QUERY:

```
SELECT *
FROM orders
ORDER BY ord_date;
```

```
mysql> SELECT *
    -> FROM orders
    -> ORDER BY ord_date;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
|  70012 |    250.45 | 2012-06-27 |        3008 |        5002 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70004 |    110.50 | 2012-08-17 |        3009 |        5003 |
|  70011 |     75.29 | 2012-08-17 |        3003 |        5007 |
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
|  70001 |    150.50 | 2012-10-05 |        3005 |        5002 |
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70010 |   1983.43 | 2012-10-10 |        3004 |        5006 |
+--------+-----------+------------+-------------+-------------+
12 rows in set (0.01 sec)
```

QUESTION 6 - SUM()

SCENARIO:
Finance department wants total revenue generated.

QUERY:

```
SELECT SUM(purch_amt) AS total_revenue
FROM orders;
```

```
mysql> SELECT SUM(purch_amt) AS total_revenue
    -> FROM orders;
+---------------+
| total_revenue |
+---------------+
|      17541.18 |
+---------------+
1 row in set (0.03 sec)
```

QUESTION 7 - AVG()

SCENARIO:
Management wants average order value.

QUERY:

```
SELECT AVG(purch_amt) AS average_order
FROM orders;
```

```
mysql> SELECT AVG(purch_amt) AS average_order
    -> FROM orders;
+---------------+
| average_order |
+---------------+
|   1461.765000 |
+---------------+
1 row in set (0.00 sec)
```

QUESTION 8 - MAX()

SCENARIO:
Find the highest order amount.

QUERY:

```
SELECT MAX(purch_amt) AS highest_order
FROM orders;
```

```
mysql> SELECT MAX(purch_amt) AS highest_order
    -> FROM orders;
+---------------+
| highest_order |
+---------------+
|       5760.00 |
+---------------+
1 row in set (0.01 sec)
```

QUESTION 9 - MIN()

SCENARIO:
Find the lowest order amount.

QUERY:

```
SELECT MIN(purch_amt) AS lowest_order
FROM orders;
```

```
mysql> SELECT MIN(purch_amt) AS lowest_order
    -> FROM orders;
+--------------+
| lowest_order |
+--------------+
|        65.26 |
+--------------+
1 row in set (0.00 sec)
```

QUESTION 10 - COUNT()

SCENARIO:
Find total number of orders.

QUERY:

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

```
mysql> SELECT COUNT(*) AS total_orders
    -> FROM orders;
+--------------+
| total_orders |
+--------------+
|           12 |
+--------------+
1 row in set (0.06 sec)
```

QUESTION 11 - GROUP BY

SCENARIO:
Sales director wants total sales generated by each salesman.

QUERY:

```
SELECT salesman_id,
       SUM(purch_amt) AS total_sales
FROM orders
GROUP BY salesman_id;
```

```
mysql> SELECT salesman_id,
    ->        SUM(purch_amt) AS total_sales
    -> FROM orders
    -> GROUP BY salesman_id;
+-------------+-------------+
| salesman_id | total_sales |
+-------------+-------------+
|        5002 |     1349.45 |
|        5001 |    11271.46 |
|        5003 |     2590.90 |
|        5005 |      270.65 |
|        5006 |     1983.43 |
|        5007 |       75.29 |
+-------------+-------------+
6 rows in set (0.01 sec)
```

QUESTION 12 - GROUP BY

SCENARIO:
Marketing wants total purchases made by each customer.

QUERY:

```
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id;
```

```
mysql> SELECT customer_id,
    ->        SUM(purch_amt) AS total_purchase
    -> FROM orders
    -> GROUP BY customer_id;
+-------------+----------------+
| customer_id | total_purchase |
+-------------+----------------+
|        3005 |        1099.00 |
|        3002 |        8870.86 |
|        3009 |        2590.90 |
|        3007 |        2400.60 |
|        3001 |         270.65 |
|        3004 |        1983.43 |
|        3003 |          75.29 |
|        3008 |         250.45 |
+-------------+----------------+
8 rows in set (0.01 sec)
```

QUESTION 13 - GROUP BY + MAX()

SCENARIO:
Find the highest order placed by each customer.

QUERY:

```
SELECT customer_id,
       MAX(purch_amt) AS highest_purchase
FROM orders
GROUP BY customer_id;
```

```
mysql> SELECT customer_id,
    ->        MAX(purch_amt) AS highest_purchase
    -> FROM orders
    -> GROUP BY customer_id;
+-------------+------------------+
| customer_id | highest_purchase |
+-------------+------------------+
|        3005 |           948.50 |
|        3002 |          5760.00 |
|        3009 |          2480.40 |
|        3007 |          2400.60 |
|        3001 |           270.65 |
|        3004 |          1983.43 |
|        3003 |            75.29 |
|        3008 |           250.45 |
+-------------+------------------+
8 rows in set (0.00 sec)
```

QUESTION 14 - GROUP BY + HAVING

SCENARIO:
Company rewards salesmen whose total sales exceed $3,000.

QUERY:

```
SELECT salesman_id,
       SUM(purch_amt) AS total_sales
FROM orders
GROUP BY salesman_id
HAVING SUM(purch_amt) > 3000;
```

```
mysql> SELECT salesman_id,
    ->        SUM(purch_amt) AS total_sales
    -> FROM orders
    -> GROUP BY salesman_id
    -> HAVING SUM(purch_amt) > 3000;
+-------------+-------------+
| salesman_id | total_sales |
+-------------+-------------+
|        5001 |    11271.46 |
+-------------+-------------+
1 row in set (0.01 sec)
```

QUESTION 15 - GROUP BY + HAVING

SCENARIO:
Find customers whose total purchases exceed $2,500.

QUERY:

```
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id
HAVING SUM(purch_amt) > 2500;
```

```
mysql> SELECT customer_id,
    ->        SUM(purch_amt) AS total_purchase
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING SUM(purch_amt) > 2500;
+-------------+----------------+
| customer_id | total_purchase |
+-------------+----------------+
|        3002 |        8870.86 |
|        3009 |        2590.90 |
+-------------+----------------+
2 rows in set (0.00 sec)
```

QUESTION 16 - GROUP BY + HAVING

SCENARIO:
Find customers who placed more than one order.

QUERY:

```
SELECT customer_id,
       COUNT(*) AS total_orders
FROM orders
GROUP BY customer_id
HAVING COUNT(*) > 1;
```

```
mysql> SELECT customer_id,
    ->        COUNT(*) AS total_orders
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING COUNT(*) > 1;
+-------------+--------------+
| customer_id | total_orders |
+-------------+--------------+
|        3005 |            2 |
|        3002 |            3 |
|        3009 |            2 |
+-------------+--------------+
3 rows in set (0.00 sec)
```

QUESTION 17 - GROUP BY + HAVING + ORDER BY

SCENARIO:
Management wants top customers ranked by spending.

QUERY:

```
SELECT customer_id,
       SUM(purch_amt) AS total_purchase
FROM orders
GROUP BY customer_id
HAVING SUM(purch_amt) > 1000
ORDER BY total_purchase DESC;
```

```
mysql> SELECT customer_id,
    ->        SUM(purch_amt) AS total_purchase
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING SUM(purch_amt) > 1000
    -> ORDER BY total_purchase DESC;
+-------------+----------------+
| customer_id | total_purchase |
+-------------+----------------+
|        3002 |        8870.86 |
|        3009 |        2590.90 |
|        3007 |        2400.60 |
|        3004 |        1983.43 |
|        3005 |        1099.00 |
+-------------+----------------+
5 rows in set (0.03 sec)
```

QUESTION 18 - BETWEEN + HAVING

SCENARIO:
Find customers whose maximum purchase is between $2,000 and $6,000.

QUERY:

```
SELECT customer_id,
       MAX(purch_amt) AS max_purchase
FROM orders
GROUP BY customer_id
HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
```

```
mysql> SELECT customer_id,
    ->        MAX(purch_amt) AS max_purchase
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
+-------------+--------------+
| customer_id | max_purchase |
+-------------+--------------+
|        3002 |      5760.00 |
|        3009 |      2480.40 |
|        3007 |      2400.60 |
+-------------+--------------+
3 rows in set (0.01 sec)
```

QUESTION 19 - COUNT + HAVING

SCENARIO:
Find salesmen who handled at least two orders.

QUERY:

```
SELECT salesman_id,
       COUNT(*) AS total_orders
FROM orders
GROUP BY salesman_id
HAVING COUNT(*) >= 2;
```

```
mysql> SELECT salesman_id,
    ->        COUNT(*) AS total_orders
    -> FROM orders
    -> GROUP BY salesman_id
    -> HAVING COUNT(*) >= 2;
+-------------+--------------+
| salesman_id | total_orders |
+-------------+--------------+
|        5002 |            3 |
|        5001 |            4 |
|        5003 |            2 |
+-------------+--------------+
3 rows in set (0.01 sec)
```

QUESTION 20 - MAX + GROUP BY + HAVING

SCENARIO:
Management wants daily highest purchases above $2,000.

QUERY:

```
SELECT ord_date,
```

```
       MAX(purch_amt) AS highest_purchase
FROM orders
GROUP BY ord_date
HAVING MAX(purch_amt) > 2000;
```

```
mysql> SELECT ord_date,
    ->        MAX(purch_amt) AS highest_purchase
    -> FROM orders
    -> GROUP BY ord_date
    -> HAVING MAX(purch_amt) > 2000;
+------------+------------------+
| ord_date   | highest_purchase |
+------------+------------------+
| 2012-10-10 |          2480.40 |
| 2012-07-27 |          2400.60 |
| 2012-09-10 |          5760.00 |
| 2012-04-25 |          3045.60 |
+------------+------------------+
4 rows in set (0.00 sec)
```